

# INTERHOSPITAL CONFERENCE

**A 4-year-old girl presented with  
fever and headache for 2 weeks**

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## Patient Identification

- ผู้ป่วยเด็กหญิง ชาวกะเหรี่ยง อายุ 4 ปี
- ภูมิลำเนา: อำเภออุ้มผาง จังหวัดตาก
- แหล่งข้อมูล: มารดา และใบส่งตัว

# Chief Complaint

มีไข้ ปวดศีรษะ 2 สัปดาห์ก่อนมาโรงพยาบาล

## History and Present illness

2 สัปดาห์ก่อนมาโรงพยาบาล ผู้ป่วยมีไข้ต่ำ ๆ (ไม่ได้วัดไข้) มีอาการปวดศีรษะ อาเจียนวันละ 3 - 4 ครั้ง ไม่มีไอ น้ำมูก เสมหะ ไม่มีปวดท้อง ถ่ายเหลว มารดาพาไปตรวจที่โรงพยาบาลชุมชน ได้ยารักษาตามอาการ กลับไปรับประทานที่บ้าน

10 วันก่อนมาโรงพยาบาล ผู้ป่วยยังมีไข้ต่ำ ๆ อาการปวดศีรษะเป็นมากขึ้น ผู้ป่วยบ่นอาการปวดบริเวณหน้าผากและกระบอกตา รับประทานอาหารได้ลดลง มีอาการอาเจียนทุกครั้งหลังรับประทานอาหาร มารดาสงเกตว่าผู้ป่วยเริ่มซึมลง เล่นน้อยลง อ่อนเพลีย หลับมากขึ้น จึงพากลับไปตรวจที่โรงพยาบาลชุมชน แห่งเดิม ได้เข้ารับการรักษาในโรงพยาบาล และให้ IV hydration

## History and Present illness

8 วันก่อนมาโรงพยาบาล ระหว่างรักษาที่โรงพยาบาลชุมชน ผู้ป่วยยังคงมีไข้ต่ำ ๆ (BT 37.8-38°C) ร่วมกับมีอาการปวดศีรษะมากขึ้น ตรวจร่างกายพบ positive neck stiffness แพทย์ได้ให้ empiric antibiotic เป็น cefotaxime IV 300 mg/kg/day แบ่งให้ทุก 8 ชั่วโมง และได้ตรวจน้ำไขสันหลัง

4 วันก่อนมาโรงพยาบาล ผู้ป่วยเริ่มมีอาการตามัว มองไม่เห็น มารดาสังเกตเห็นผู้ป่วยมีอาการตาซ้ายเข้เข้าด้านใน อาการปวดศีรษะไม่ดีขึ้น แพทย์ได้ทำการตรวจ CT brain (non-contrast) เจาะตรวจน้ำไขสันหลังซ้ำ และส่งตัวมารับการรักษาต่อที่โรงพยาบาลมหาวิทยาลัยเชียงใหม่

# Investigations แรกวันที่รพ.ชุมชน

| CBC                              | Result  |
|----------------------------------|---------|
| Hb (g/dL)                        | 12.2    |
| Hct (%)                          | 37      |
| WBC (cell/mm <sup>3</sup> )      | 13,580  |
| Neutrophil(%)                    | 60.8    |
| Lymphocyte(%)                    | 22.6    |
| Eosinophil(%)                    | 10.4    |
| Monocyte (%)                     | 5.9     |
| Platelet (cell/mm <sup>3</sup> ) | 319,000 |
| AEC                              | 1,412   |

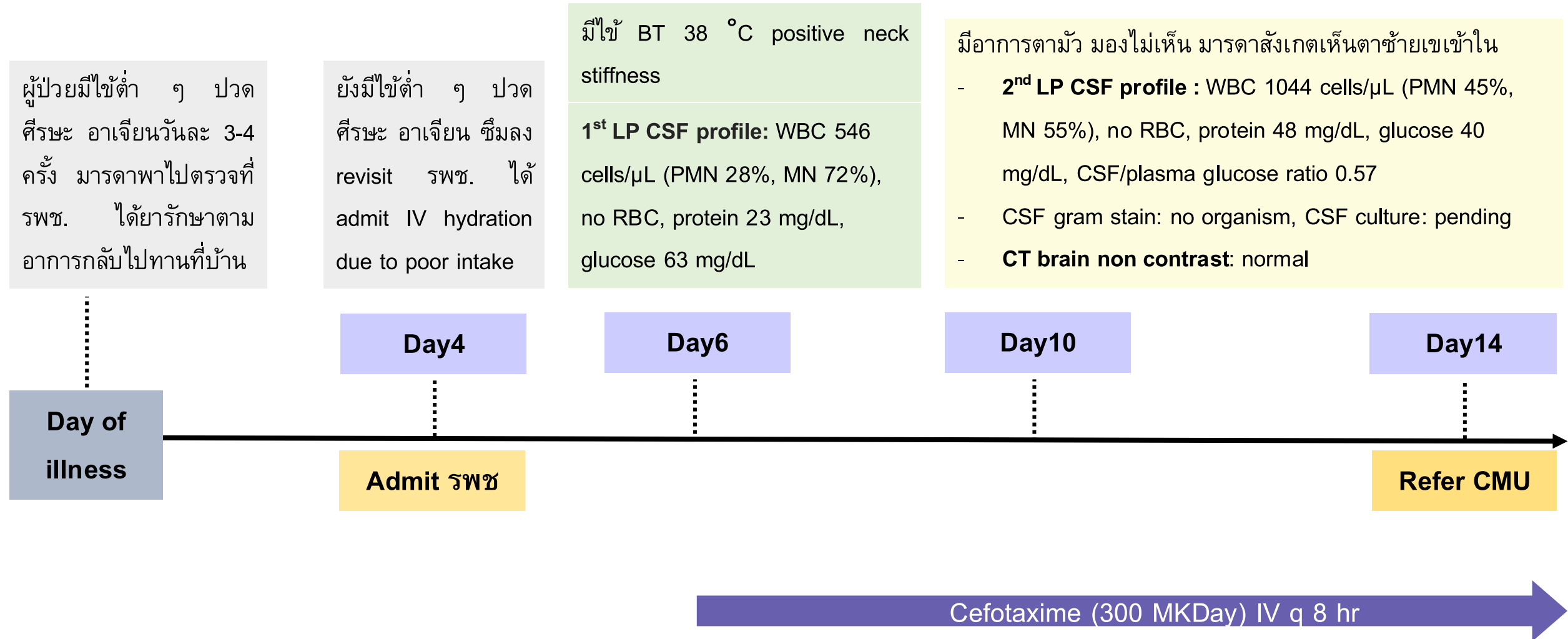
| CSF profile                         | Results (1 <sup>st</sup> )        | Results (2 <sup>nd</sup> )         |
|-------------------------------------|-----------------------------------|------------------------------------|
| Cell count and cell diff (cells/uL) | WBC 546 (PMN 28%, MN 72%), no RBC | WBC 1044 (PMN 45%, MN 55%), no RBC |
| CSF protein (mg/dL)                 | 23                                | 48                                 |
| CSF glucose (mg/dL)                 | 63                                | 40                                 |
| CSF/plasma glucose ratio            | 0.75                              | 0.57                               |
| CSF gram stain                      | no organism                       | no organism                        |
| CSF C/S                             | no growth                         | no growth                          |
| OP (mmHg)                           | not done                          | cannot evaluate                    |
| CP (mmHg)                           | not done                          | 27                                 |

## CT brain non contrast (รพ.ชุมชน)

- No intracranial hemorrhage or large infarction
- No shifting of midline structure or brain herniation
- No dilated ventricular systems
- Both orbits, PNS and bilateral mastoid air cell appear unremarkable
- Bony carvarium are intact

**Impression:** no intracranial hemorrhage or recent large infarction









## Past History

- คลอดครบกำหนด ไม่มีภาวะแทรกซ้อน
- มารดาแจ้งว่าได้รับวัคซีนแรกเกิดและวัคซีนตามอายุครบ ไม่ได้วัคซีนเสริม (ไม่ได้นำสมุดวัคซีนมา)
- ปฏิเสธโรคประจำตัว หรือยาที่รับประทานประจำ
- ก่อนหน้านี้สุขภาพแข็งแรงดี
- ไม่มีประวัติโรคมะเร็งหรือโรคทางพันธุกรรมในครอบครัว

# Physical Examinations at CMU

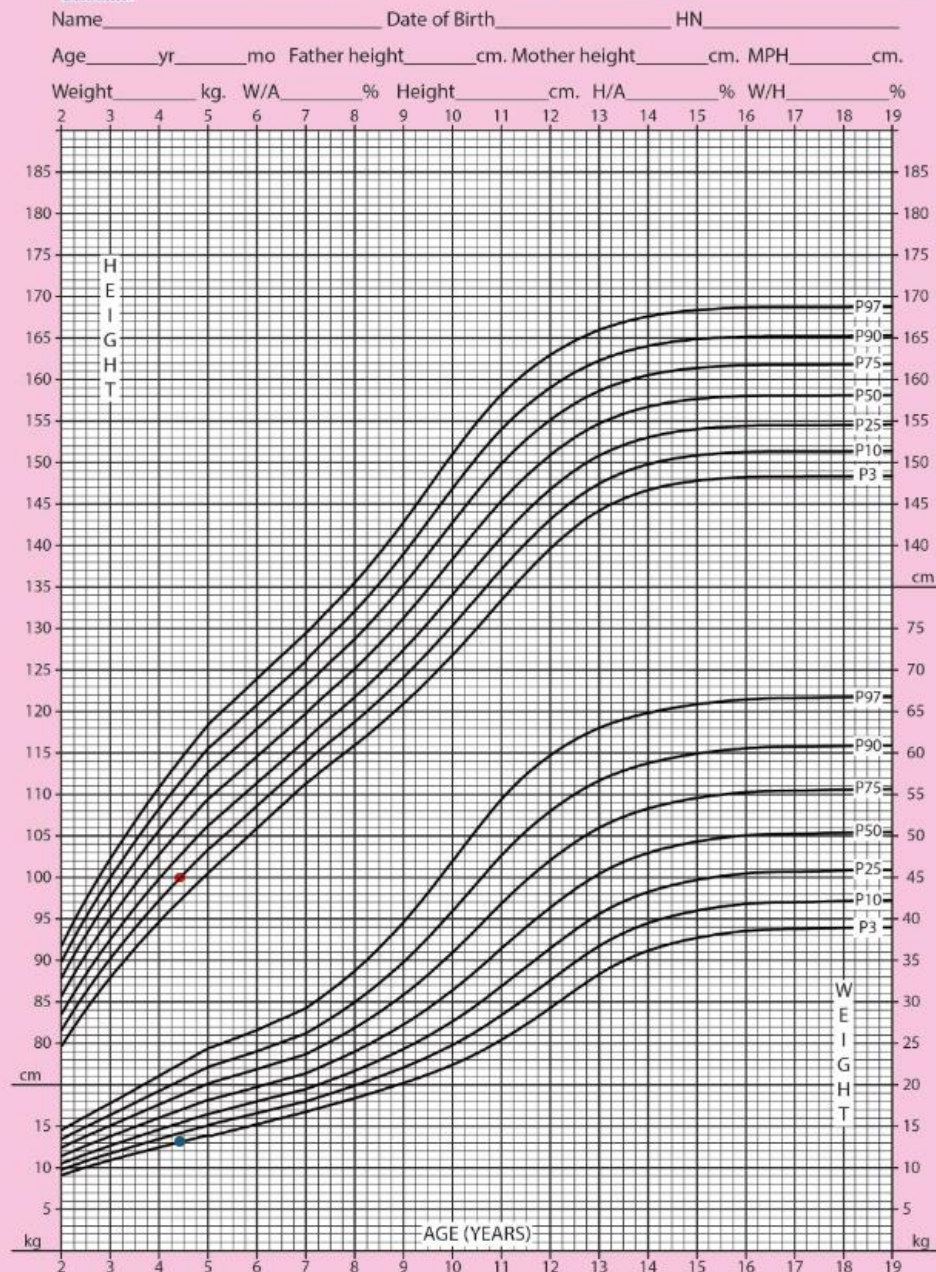
- Vital signs: **BT 38.2°C**, RR 24/min, BP 116/82 mmHg, HR 132/min, SpO2 98% (room air)
- General appearance: a young girl, **irritable and poor co-operation**
- HEENT: No pale conjunctiva, no icteric sclera, no injected pharynx, no palpable cervical lymph nodes
- Lungs: clear and equal breath sound both lungs
- Heart: normal S1 & S2, regular rhythm, no murmur
- Abdomen: no distension, soft, no point of tender, no hepatosplenomegaly
- Extremities: seen BCG scar at left arm, no joint swelling, no rash, CRT<2 secs, pulse full 2+, no deformity

# Physical Examinations at CMU

- **Neurological examination:**
  - E4V5M6, pupil 3 mm RTLBE, **not fix and follow objects, no eye contact,**  
**esotropia both eyes in resting position**
  - No facial palsy, **limited lateral rectus both eyes**
  - Sensation: cannot evaluate
  - Motor power: at least grade IV all, normotonia, reflex 2+ all
  - **Stiffness of neck: positive**
  - Clonus: negative both feet
  - Babinski's reflex: plantarflexion both feet
- Eye ground: **papilledema both eyes**

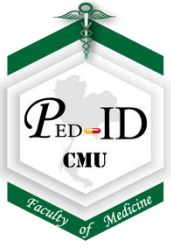






## Growth Curve

- BW 13.2 kg (P3)
- Height 100 cm (P10)

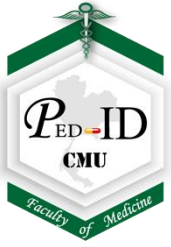


# History taking?

## Additional History

- ปฏิเสธประวัติ contact TB, ที่บ้านอาศัยอยู่กัน 5 คน ปู่ พ่อ แม่ พี่ชาย ทุกคนแข็งแรงดี ไม่มีใครไอเรื้อรัง
- รับประทานผักสดเป็นประจำ เช่น กะหล่ำปลี แตงกวา ผักสลัด
- เคยทานปูนาต้ม/ทอด ทานหอยเชอรี่ต้ม ทานปลาน้ำจืดทอด/ต้ม
- บริเวณแถวบ้านเป็นดินโคลน ผู้ป่วยชอบเดินเล่นเท้าเปล่า





# Problem lists and Differential Diagnosis ?

## Problem Lists

A 4-year-old girl presented with low grade fever with progressive headache for 2 weeks

- Subacute meningitis with visual loss with strabismus with cranial nerve VI palsy with increased intracranial pressure
- Eosinophilia
- Malnutrition
- History of ingesting paratenic hosts (rice-field crabs and snails)

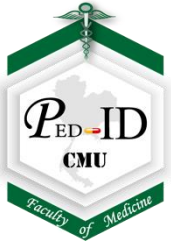
# Differential Diagnosis

## Infectious cause

- **Parasite**
  - Angiostrongyliasis
  - Gnathostomiasis
  - Cysticercosis
- **Mycobacterial**
  - *Mycobacterium tuberculosis*
- **Fungus**
  - Cryptococcosis
- **Bacterial** (with complication)
  - *Streptococcus pneumoniae*

## Non-infectious cause

- **Demyelinating**
  - Acute disseminated encephalomyelitis (ADEM)
- **Autoimmune**
  - Neuromyelitis optica (NMO)



# Investigations ?

# CBC

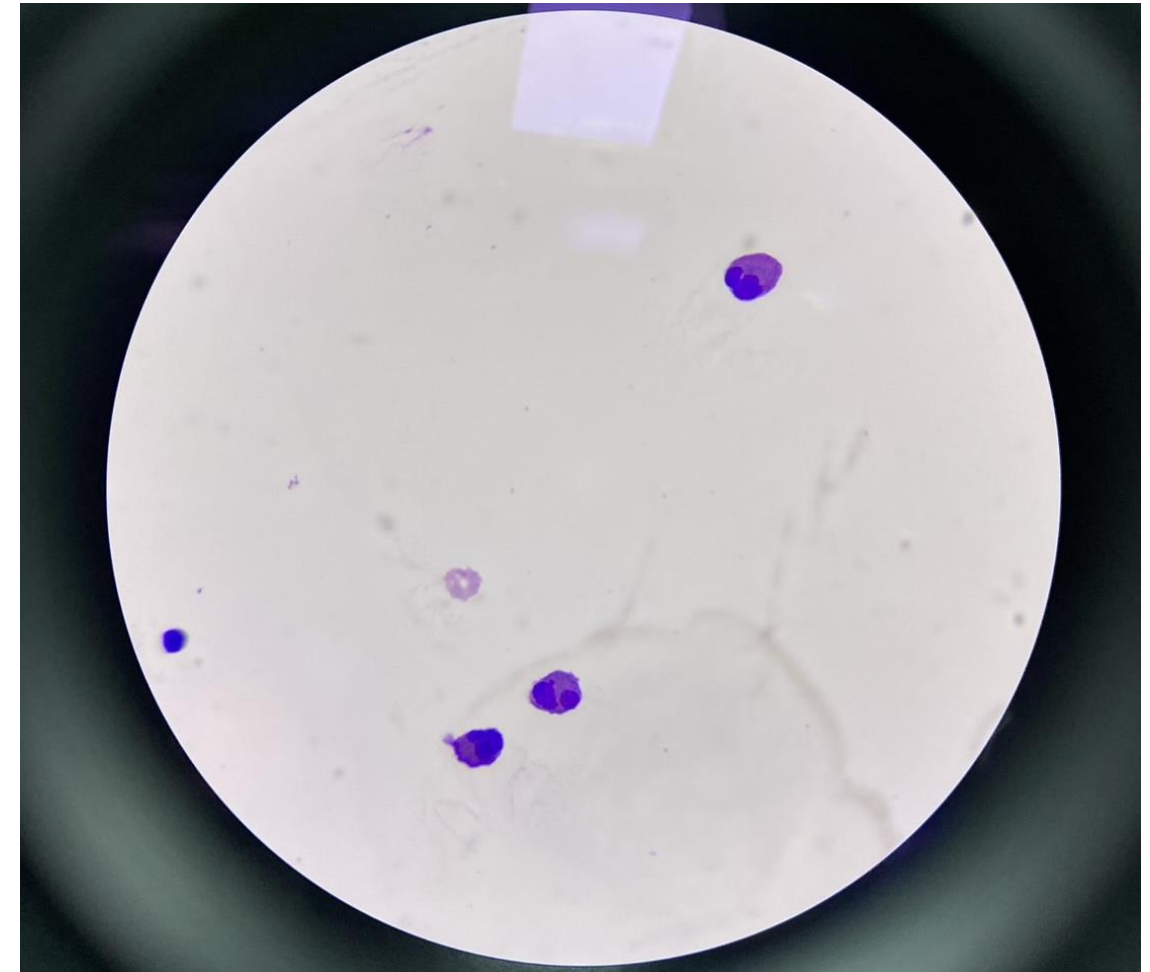
## At CMU

| CBC                   |         |
|-----------------------|---------|
| Hb (g/dL)             | 12.1    |
| Hct (%)               | 37.2    |
| WBC (cell/cu.mm)      | 13,020  |
| Neutrophil (%)        | 43.8    |
| Lymphocyte (%)        | 32.6    |
| Monocyte (%)          | 4.3     |
| Eosinophil (%)        | 19.1    |
| Basophil (%)          | 0.3     |
| ANC                   | 5,690   |
| AEC                   | 4,250   |
| Platelet (cell/cu.mm) | 483,000 |
| MCV (fl)              | 79      |



## CSF Profile at CMU

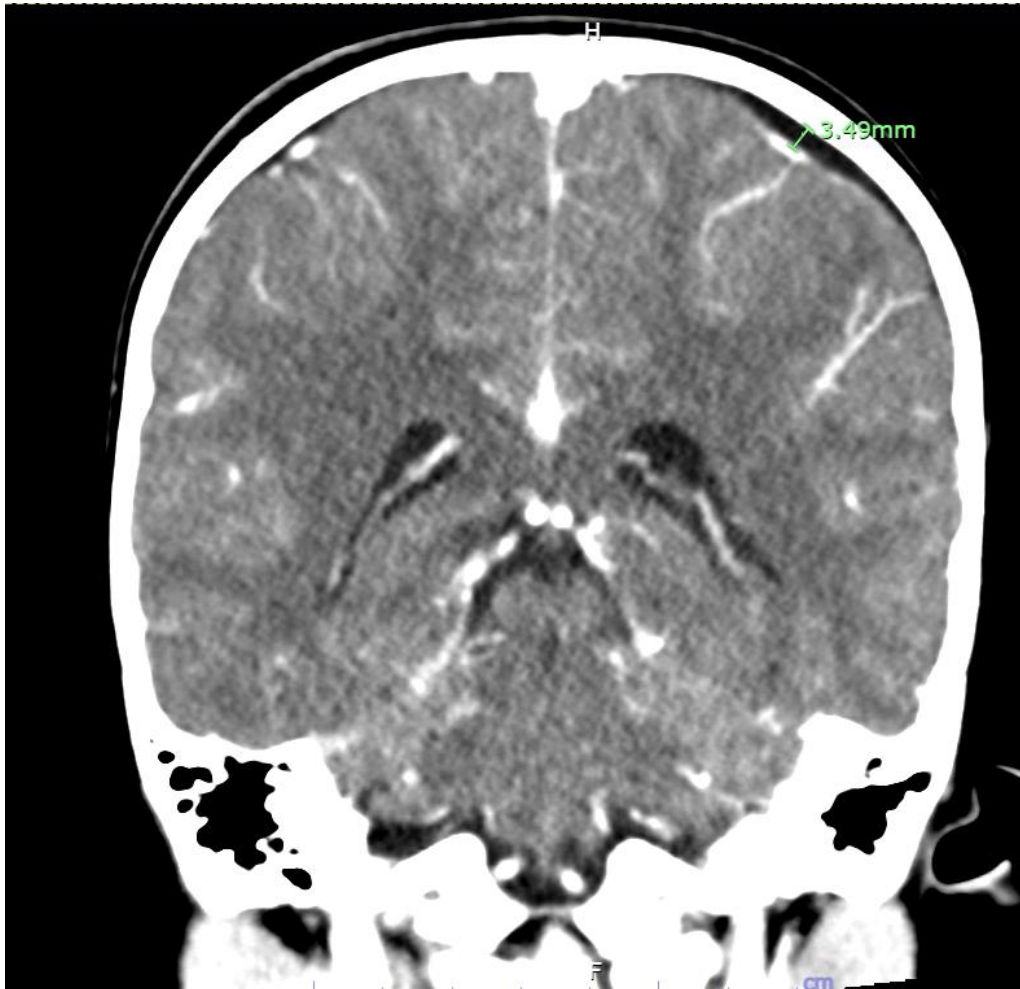
| CSF profile                         | Results (1 <sup>st</sup> )                |
|-------------------------------------|-------------------------------------------|
| Color                               | Clear                                     |
| Cell count and cell diff (cells/uL) | WBC 1,151 (PMN 56% MN 44% - Eo 40%) RBC 0 |
| CSF protein (mg/dL)                 | 60                                        |
| CSF glucose (mg/dL)                 | 37                                        |
| CSF/plasma glucose ratio            | 0.3                                       |
| CSF gram stain                      | no organism                               |
| CSF C/S                             | no growth in 2 days                       |
| OP (mmHg)                           | 22                                        |
| CP (mmHg)                           | 18                                        |



## CSF Work up

| CSF                                   | Results         |
|---------------------------------------|-----------------|
| AFB stain                             | not found       |
| Gene-XpertMTB/RIF ultra               | not detected    |
| Cryptococcus Ag                       | negative        |
| Angiostrongylus antibody (ELISA)      | Weakly positive |
| Angiostrongylus antibody (Immunoblot) | Positive        |
| Cysticercosis antibody (ELISA)        | Weakly positive |
| Gnathostomiasis antibody (ELISA)      | negative        |
| Trichinosis antibody (ELISA)          | negative        |

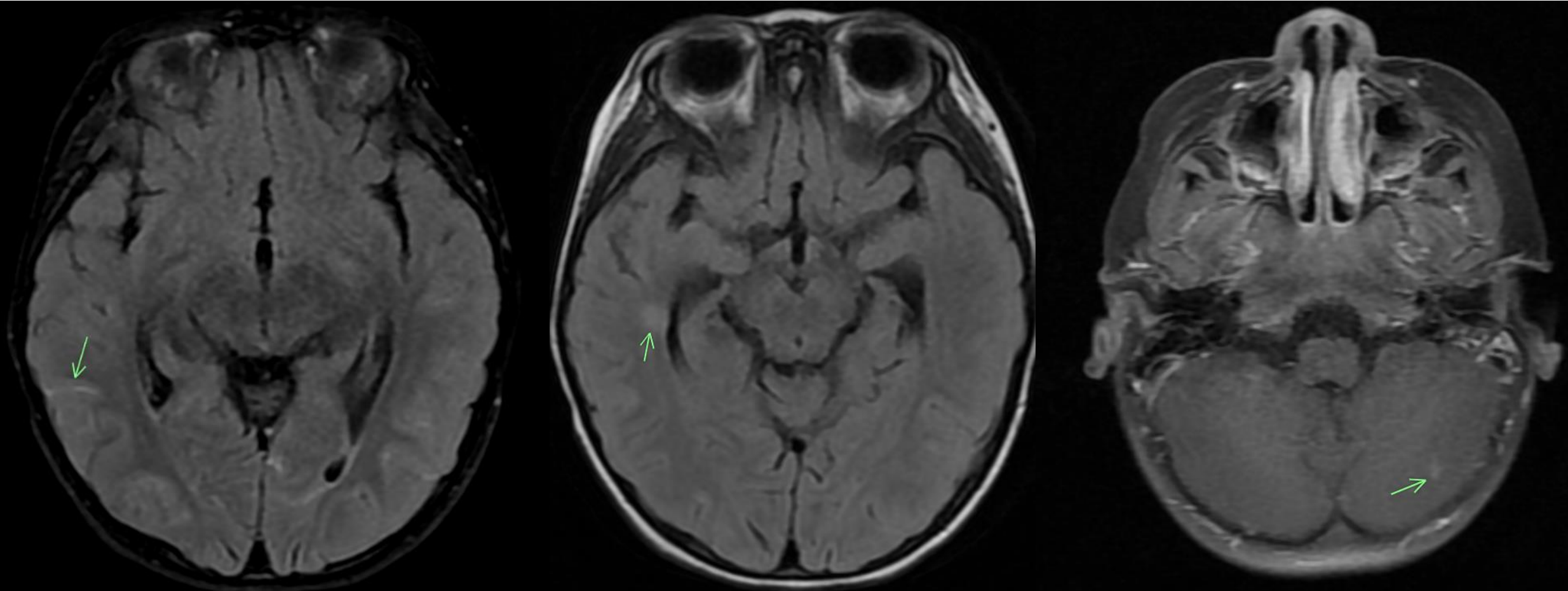
## CT Brain with Contrast at CMU



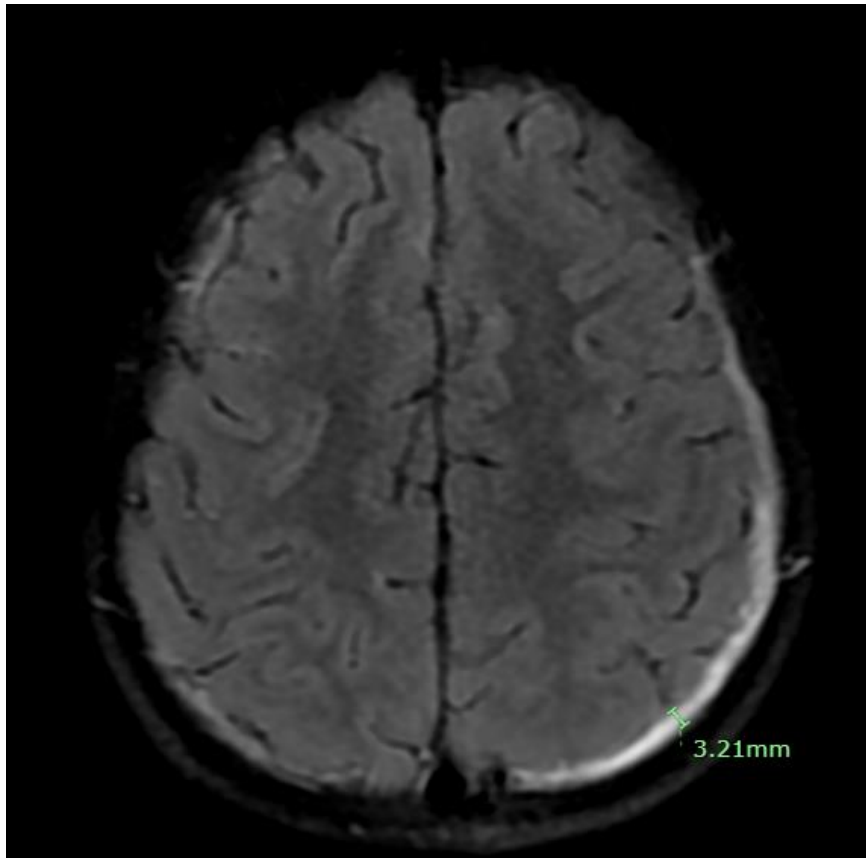
### Impression:

- Thin hypodense subdural collection along left frontoparietal convexities, measuring up to 0.3 cm
- Mild degree of communicating hydrocephalus
- No gross enhancing lesions or leptomeningeal enhancement
- No evidence of brain abscess

## MRI Brain with Orbit



## MRI Brain with Orbit



# MRI Brain with Orbit

- Again seen thin subdural effusion along the right parietal and left fronto-parietal convexities, measuring up to 0.3-cm
- Multifocal leptomeningeal enhancement; predominant at bilateral parieto-occipital cortical sulci and cerebellar folia
- A small non-enhancing T2 hyperintense signal lesion with track-like dark signal of susceptibility artifacts in SWI involving periventricular white matter lateral to temporal horn of the right lateral ventricle
- A tiny nodular enhancing lesion with central blooming artifact in left cerebellar hemisphere
- Stable degree of mild communicating hydrocephalus
- Intraocular protrusion of bilateral optic nerve heads with flattened posterior sclera, and prominent subarachnoid space around optic nerves. These findings are suggestive of increase intracranial pressure

**Impression: evidence of eosinophilic meningitis as correlate with CSF examination, radiologic evidence of increased intracranial pressure, stable degree of hydrocephalus**

## Work up for Parasite

| Blood for                           | Results         |
|-------------------------------------|-----------------|
| Angiostrongylus antibody (ELISA)    | negative        |
| Angiostrongylus antibody (immublot) | Weakly positive |
| Cysticercosis antibody (ELISA)      | negative        |

| stool                   | Results   |
|-------------------------|-----------|
| Stool conc for parasite | not found |



# CSF for viral meningitis panel multiplex RT-PCR

| RT-PCR                              | Results      |
|-------------------------------------|--------------|
| Human adenovirus                    | not detected |
| Herpes simplex virus type 1 (HSV 1) | not detected |
| Herpes simplex virus type 2 (HSV 2) | not detected |
| Varicella zoster virus (VZV)        | not detected |
| Ebstein-Barr's virus (EBV)          | not detected |
| Cytomegalovirus (CMV)               | not detected |
| Human herpesvirus 6 (HHV6)          | not detected |
| Human herpesvirus 7 (HHV7)          | not detected |
| Human parvovirus B19                | not detected |
| Human enterovirus                   | not detected |
| Human paraechovirus                 | not detected |

## CXR PA Upright and Lateral



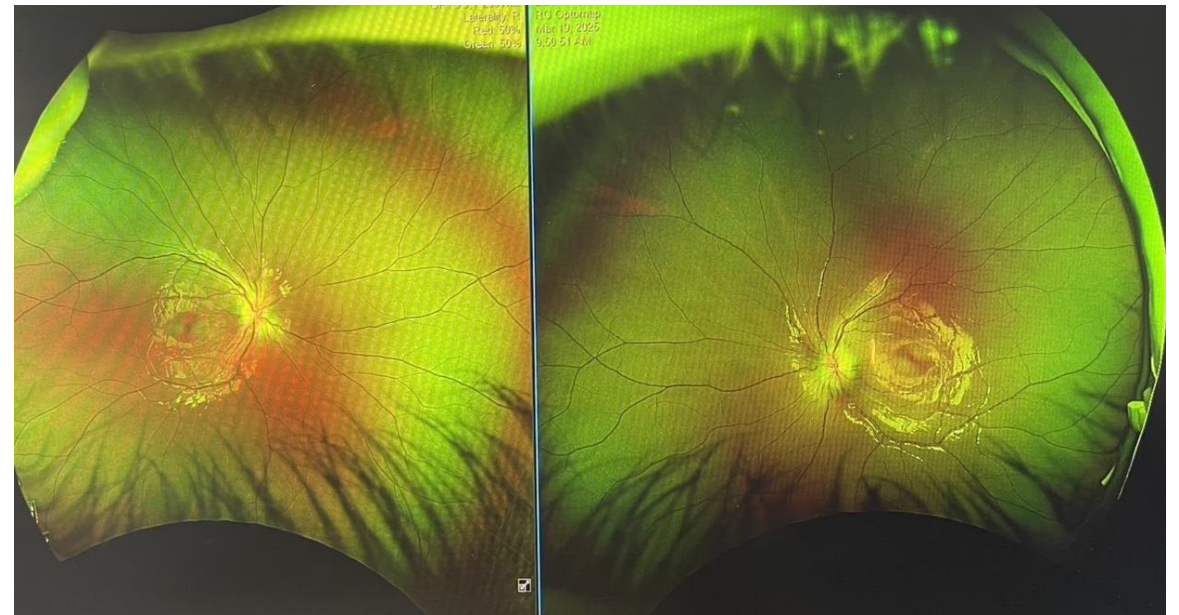
## Consult Eye for Evaluation

VA: at least perception of light, not fix and follow

RAPD: negative

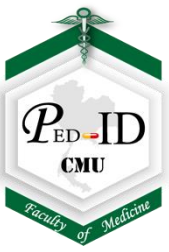
Fundoscope: **blurred optic disc in both eyes**,  
no hmg/exudate, no vasculitis, no parasitic  
lesion on fundus

**Impression: papilledema both eyes**



## Anti-HIV

- Negative



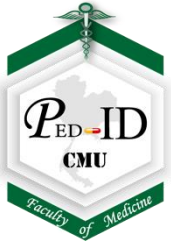
# Provisional Diagnosis ?

## Presumptive Diagnosis

### Eosinophilic meningitis caused by *Angiostrongylus cantonensis*

#### Supporting reasons:

- Common cause of eosinophilic meningitis in Thailand
- Ocular involvement
- History of exposure to intermediate hosts
- Confirmed by CSF immunology both ELISA & immunoblot



# Management ?

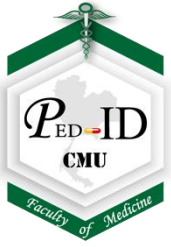


# Management

- Repeat lumbar puncture to release pressure
- Start acetazolamide for hydrocephalus 5 mg/kg/day
- Start prednisolone 1 mg/kg/day total 14 days
- Start albendazole oral for 14 days
- Plan F/U MRI brain next 3 months

# Progression

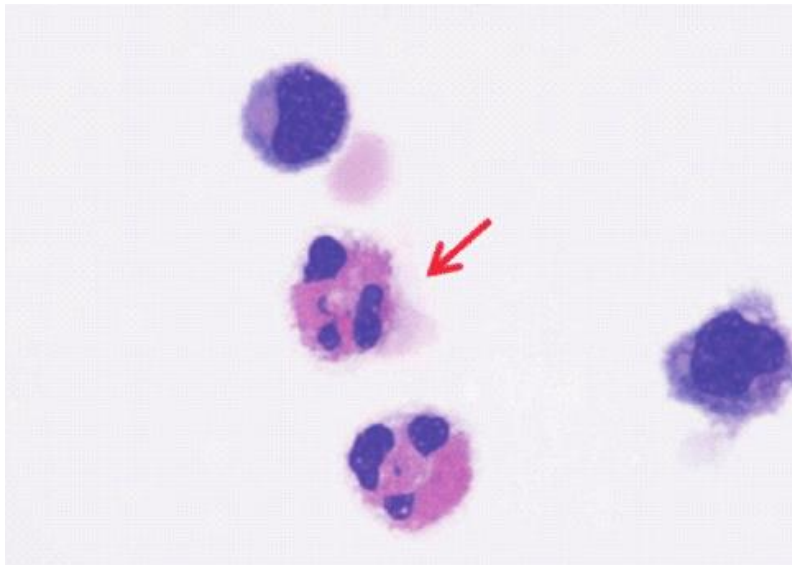
| CSF profile                         | DOA 1                                    | DOA 3                                    | DOA 10                                     |
|-------------------------------------|------------------------------------------|------------------------------------------|--------------------------------------------|
| Cell count and cell diff (cells/uL) | WBC 1151 (PMN 56%<br>MN 4% Eo 40%) RBC 0 | WBC 917 (PMN 54% MN<br>6% Eo 40%) RBC 63 | WBC 657 (PMN 43% MN<br>30% Eo 27%) RBC 333 |
| CSF protein (mg/dL)                 | 60                                       | 68                                       | 82                                         |
| CSF glucose (mg/dL)                 | 37                                       | 36                                       | 32                                         |
| CSF/plasma glucose ratio            | 0.3                                      | 0.3                                      | 0.38                                       |
| CSF gram stain                      | no organism                              | no organism                              | no organism                                |
| CSF C/S                             | no growth in 2 days                      | no growth in 2 days                      | no growth in 2 days                        |
| OP (mmHg)                           | 22                                       | 60                                       | 30                                         |
| CP (mmHg)                           | 18                                       | 40                                       | 17                                         |
| ELISA for Angiostrongylus           | Weakly positive                          | Weakly positive                          | Weakly positive                            |
| ELISA for cysticercosis             | Weakly positive                          | negative                                 | negative                                   |



# Eosinophilic Meningitis

# Etiology of Eosinophilic Meningitis

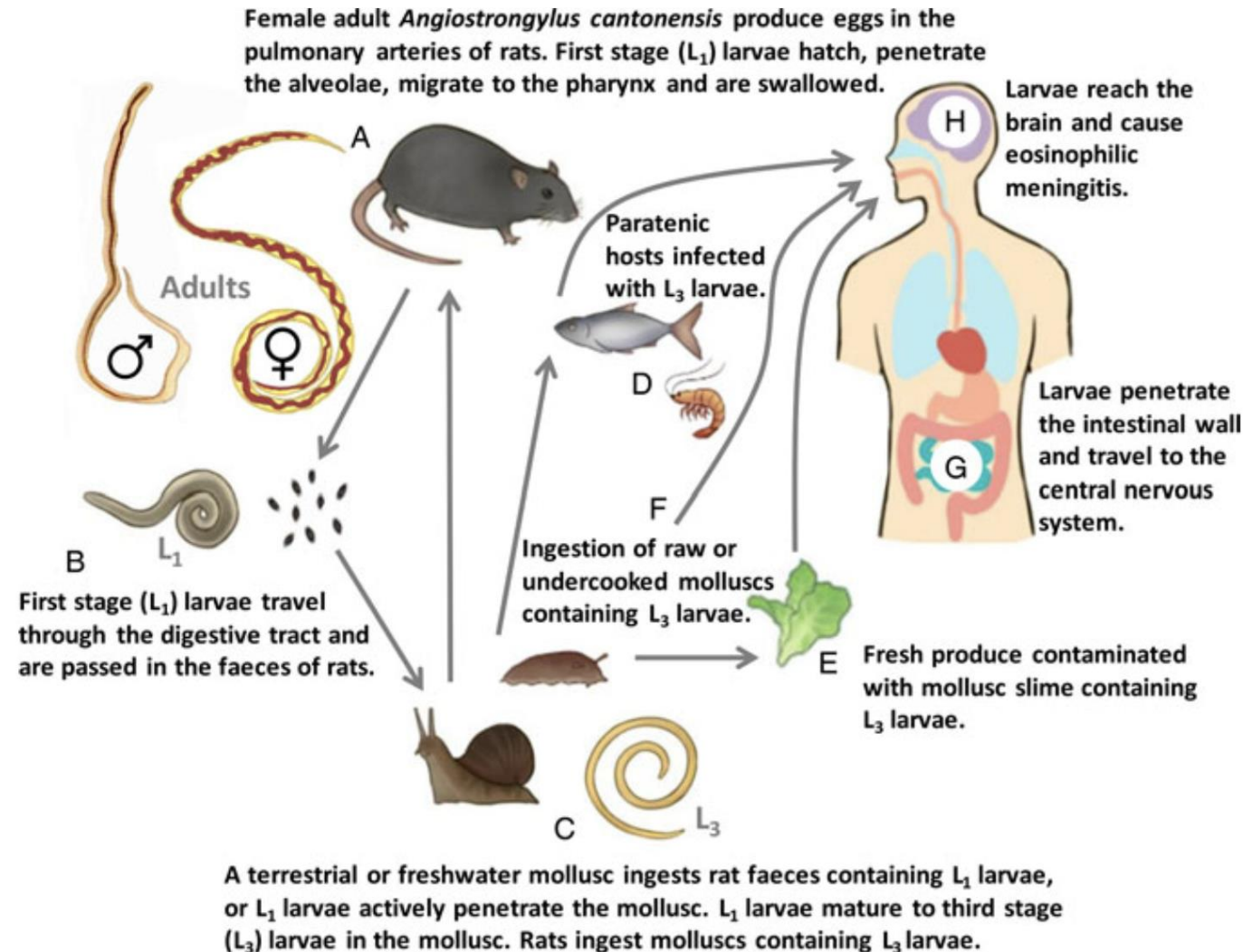
**Definition:** identification of an absolute count of 10 eosinophils per uL or more than 10% of total white blood cells in the CSF



| Infectious                                                                                                                                                                                                                                          | Noninfectious                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Parasitic</b> <ul style="list-style-type: none"> <li>- <i>Angiostrongylus cantonensis</i></li> <li>- <i>Gnathostoma spinigerum</i></li> <li>- Neurocysticercosis</li> <li>- Cerebral toxocariasis</li> <li>- Cerebral schistosomiasis</li> </ul> | <b>Malignancy</b> <ul style="list-style-type: none"> <li>- Hogkin's disease</li> <li>- Non hodgkin's lymphoma</li> </ul>                                   |
| <b>Fungus</b> <ul style="list-style-type: none"> <li>- Cryptococcosis</li> </ul>                                                                                                                                                                    | <b>Medication</b> <ul style="list-style-type: none"> <li>- Ciprofloxacin</li> <li>- Ibuprofen</li> <li>- Intraventricular vancomycin/gentamycin</li> </ul> |
| Others: mycobacterium tuberculosis                                                                                                                                                                                                                  | Hypereosinophilic syndrome                                                                                                                                 |

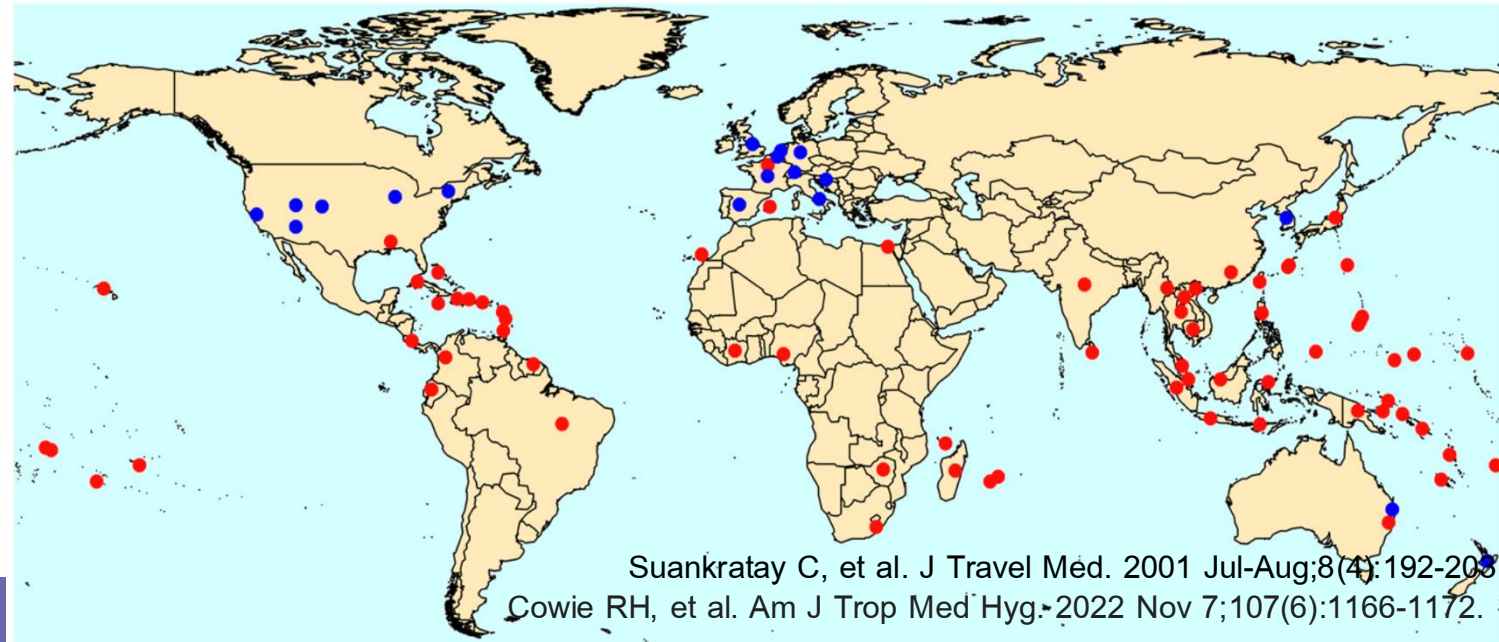
# Angiostrongylus cantonensis

- Commonly known as the **"rat lungworm"**
- Rats are the definitive hosts and the primary transmitters, **while snails and slugs act as intermediate hosts**
- Neurotropic, migrating to neural tissue after infection and resulting in three syndromes:
  - **Eosinophilic meningitis**
  - Eosinophilic meningoencephalitis
  - Ocular angiostrongyliasis: rare but can cause permanent visual loss



## Global Distribution of Neuroangiostrongyliasis

- *Angiostrongylus* spp. are endemic to the Asia-Pacific regions, with the **highest incidence in Thailand and Malaysia**, additional areas of inhabitation include the Pacific basin, China, Vietnam, the Philippines, Taiwan, and Hawaii
- Travel-related exposures have been attributed to sporadic cases detected in Europe and the United States



# Clinical Manifestations

- Symptoms began a median of 11 days (range 1 week to 1 month) after consumption of intermediate hosts
- **Headache is the most common presentation, often progressive and severe**
- Other signs/symptoms: photophobia or visual disturbance, fatigue, vomiting, paresthesia, hyperesthesia, neck stiffness, sixth or seventh cranial nerve palsy and papilledema



## Clinical Manifestations of Eosinophilic Meningitis Due to Infection with *Angiostrongylus cantonensis* in Children

- The study reviewed 19 cases of EM in children under 15 years old at Srinagarind Hospital, Khon Kaen University, Thailand
- All patients presented with severe headaches
- One-third exhibited papilledema or cranial nerve palsies
- The clinical manifestations in children differed from those typically observed in adults, with fever, nausea, vomiting, hepatomegaly, neck stiffness, and cranial nerve palsies being more prevalent in pediatric cases

### Clinical manifestation of eosinophilic meningitis in children and adults

| Variables                        | Children (N=19) | Adults (N=80) |
|----------------------------------|-----------------|---------------|
| Age, years (range)               | 12 (4-14)       | 33.5 (15-70)  |
| Male gender, N (%)               | 15 (78.9)       | 60 (75.0)     |
| Summer season (%)                | 4 (21.1)        | 12 (15.0)     |
| Snail ingestion, N (%)           | 13 (68.4)       | 80 (100)      |
| Incubation period, days (range)  | 22 (1-30)       | 20 (1-90)     |
| Headache, N (%)                  | 19 (100)        | 80 (100)      |
| Nausea or vomiting, N (%)        | 12 (63.2)       | 31 (38.8)     |
| Fever, N (%)                     | 15 (78.9)       | 8 (10)        |
| Hepatomegaly, N (%)              | 2 (10.5)        | 0 (0)         |
| 6th cranial nerve palsy, N (%)   | 5 (26.3)        | 3 (3.8)       |
| 7th cranial nerve palsy, N (%)   | 1 (5.3)         | 2 (2.5)       |
| Abnormal cerebellar signs, N (%) | 2 (10.5)        | 0 (0)         |
| Hyperesthesia, N (%)             | 0               | 9 (11.3)      |
| Neck stiffness, N (%)            | 13 (68.4)       | 38 (47.5)     |
| Papilledema, N (%)               | 6 (31.6)        | 2 (2.5)       |

# Diagnosis

- **Definitive diagnosis** is made by detection of *A.cantonensis* larvae in the CSF or eye chamber. However, the detection rate is frequently low
- **Presumptive diagnosis** based on the following:
  - Characteristic symptoms and signs + **exposure history**
  - Laboratory diagnosis: eosinophilic meningitis (EOM), Immunologic test
  - Imaging: CT and MRI findings are relatively nonspecific and are supportive rather than diagnostic

## Exposure history

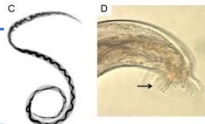
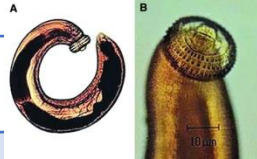


# CSF Analysis

- The CSF is usually clear or cloudy "coconut juice" appearance
- **CSF eosinophilic pleocytosis**
- The CSF protein concentration is elevated, whereas the glucose level is normal or slightly reduced
- Opening CSF pressures are elevated in nearly 40% of cases
- Immunologic techniques have been developed to confirm a presumptive diagnosis
  - ELISA and Western blot testing have been described to detect antibodies against *A. cantonensis*
  - The reaction of serum antibodies to the 31-kd and 29-kd antigens of *A. cantonensis* by western blot analysis provides a relatively specific immunodiagnosis



# Clinical features to differentiate the two main causes of eosinophilic meningitis

| Features                   | Angiostrongyliasis  | Gnathostomiasis  |
|----------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Presenting symptoms        | Acute severe headache                                                                                  | Motor weakness                                                                                      |
| Migratory swelling         | none                                                                                                   | yes                                                                                                 |
| Pain                       | With focal numbness                                                                                    | Along nerve root                                                                                    |
| Peripheral eosinophilia    | yes                                                                                                    | yes                                                                                                 |
| CSF appearance             | Coconut juice                                                                                          | Nontraumatic bloody                                                                                 |
| Brain imaging              | No pathognomonic sign                                                                                  | SAH or unusual site ICH                                                                             |
| History of larval exposure | Uncooked snail or slugs                                                                                | Uncooked fish, poultry                                                                              |
| Diagnostic immunoblot band | 29 or 31 kDa                                                                                           | 21 or 24 kDa                                                                                        |

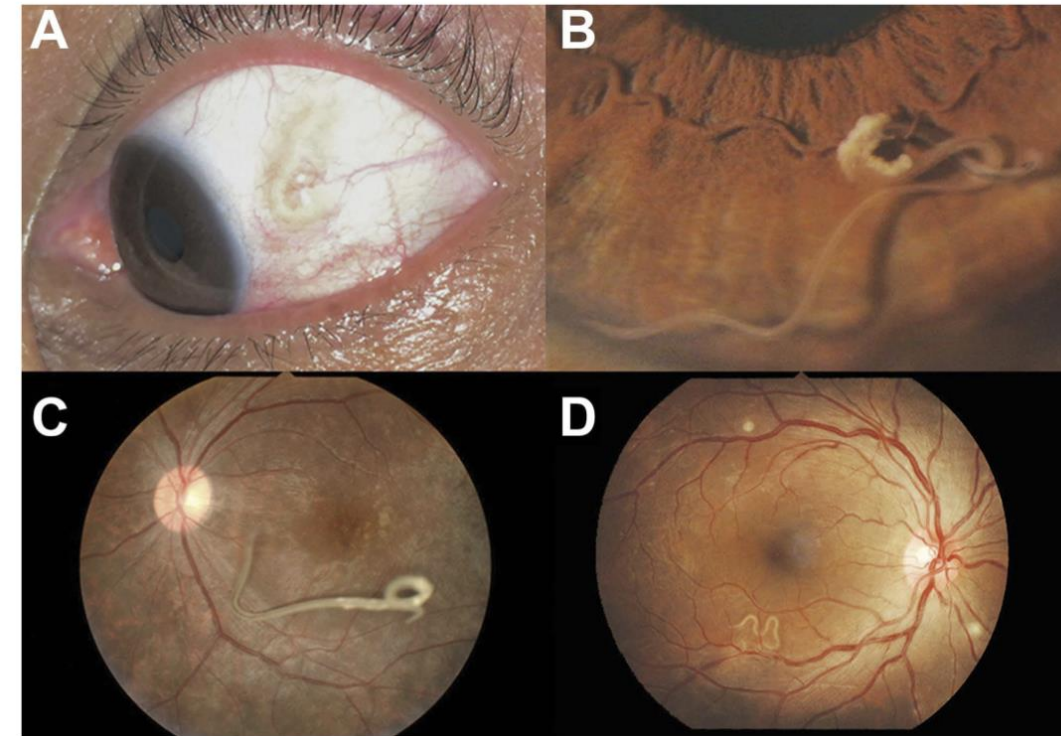
# Treatment

- **Supportive treatment**
  - Analgesic drugs should be administered for pain relief
  - Repeated lumbar punctures as needed to decrease intracranial pressure, thereby reducing pain symptoms
- **Corticosteroid**
  - A two-week course of corticosteroids is recommended in the absence of an obvious contraindication
- **Antihelminthic** – increasing number of guidelines recommend albendazole-corticosteroid cotherapy

# Literature Reviews

## Ocular angiostrongyliasis in Thailand: a retrospective analysis over two decades

- Single-center retrospective study, between 1995 and 2017
- At Srinagarind Hospital, Khon Kaen, Thailand
- A total of 18 patients were diagnosed through identification of the *Angiostrongylus cantonensis* parasite within the eye
- All patients had a history of raw food ingestion, such as snails
- Most of the patients presented with blurred vision
- 13 cases had best-corrected visual acuity of 2/60 or worse
- Multiple treatment modalities were used: focal laser, surgery, antihelminthic drugs, and steroid - majority of cases (70.6%) **did not have visual improvement**





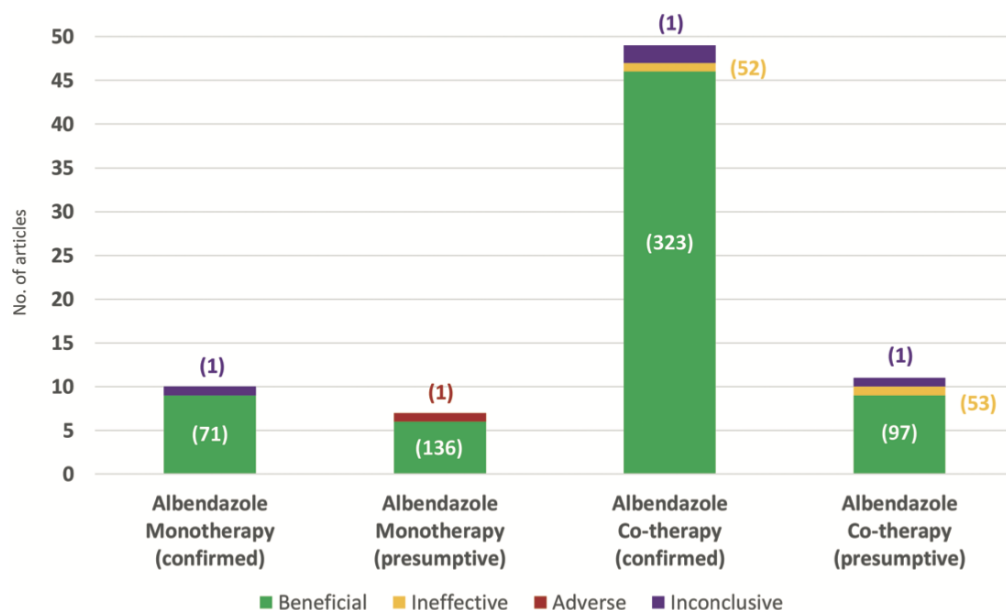
# Literature Reviews

## **Corticosteroids for Parasitic Eosinophilic Meningitis: Systematic Review by Thanaviratananich et al., 2015**

- Found only one RCT, included 129 patients (63 in the treatment group, prednisolone 60 mg/day for 2 weeks)
- The outcomes favored the treatment group
- **Shortening the median time to resolution of headaches** ( 5 vs 13 days in the control group,  $P < 0.0001$ )
- Smaller numbers of patients who still had headaches after a 2 weeks of treatment (9.1% vs 45.5%,  $P < 0.0001$ )
- Smaller numbers of patients who needed repeat lumbar puncture (12.7% vs 40%,  $P = 0.002$ )
- Reduction in the median time of analgesic use (10.5 versus 25.0,  $P = 0.038$ )
- There were **no reported adverse effects** from prednisolone in the treatment group



## Clinical Efficacy and Safety of Albendazole and Other Benzimidazole Anthelmintics



### A Systematic Analysis of Clinical Reports and Animal Studies

- 144 articles (104 clinical reports & 40 animal studies), covering 1034 patients and 2561 animals, data from 18 laboratories and 24 countries

### Key findings:

- 90.5% found benzimidazoles beneficial (885 patients, 2530 animals)
- 4.3% found them safe but ineffective (73 patients, 3 animals)
- 5% reported adverse outcomes (7 patients, 28 animals).

**Conclusion:** strong evidence supports the safe and effective use of benzimidazoles, particularly albendazole + corticosteroids, for managing neuroangiostrongyliasis, particularly early in the infection

- Duration of treatment approximately 2-2.5 weeks
- Mean time of initiation of treatment 3.5-4.5 weeks after symptoms onset

## Take home messages

- *Angiostrongylus cantonensis* is a leading cause of eosinophilic meningitis in Thailand
- Early recognition is crucial, particularly in children with exposure to paratenic hosts
- Automated systems may not always report eosinophil counts, so manual review is essential when clinically suspected
- Steroid treatment is effective in reducing inflammation and symptoms
- Early antihelminthic treatment is beneficial and supported by current evidence
- Prevention: avoid raw/undercooked snails, slugs, prawns; practice good food hygiene and handwashing

**Thank You**